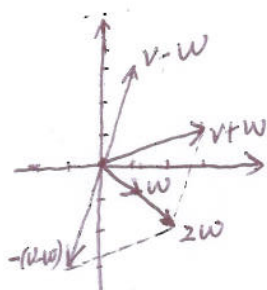
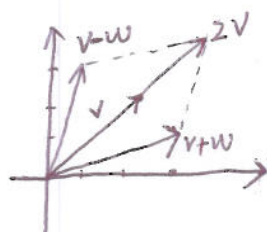


§1.1.

P3. Solution: $v+w = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, v-w = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$$\Rightarrow 2v = (v+w) + (v-w) = \begin{bmatrix} 4 \\ 4 \end{bmatrix} \Rightarrow v = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$2w = (v+w) - (v-w) = \begin{bmatrix} 2 \\ -2 \end{bmatrix} \Rightarrow w = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$



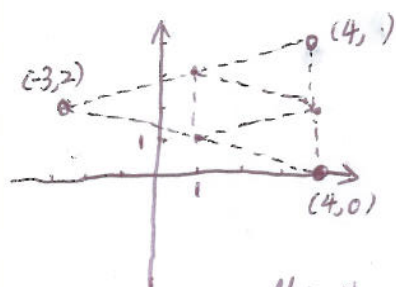
P6. Solution:

$$cv + dw = c \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} c \\ -2c \\ c \end{bmatrix} + \begin{bmatrix} 0 \\ d \\ -d \end{bmatrix} = \begin{bmatrix} c \\ -2c+d \\ c-d \end{bmatrix}$$

$$cv + dw = \begin{bmatrix} 4 \\ 2 \\ -6 \end{bmatrix} \Rightarrow$$

$$\begin{cases} c = 4 \\ -2c + d = 2 \\ c - d = -6 \end{cases} \Rightarrow \begin{cases} c = 4 \\ d = 10 \end{cases}$$

P9. Solution:

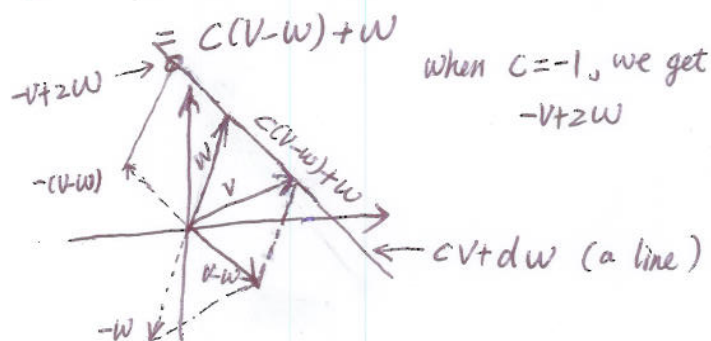


There are totally three choices
(4,), (4,0), (-3,2)

P16. Solution:

$$c+d=1$$

$$cv + dw = cv + (1-c)w$$



§1.2

P3. Solution

$$u = \begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}, v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, w = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

Normalize $v \Rightarrow$

$$\frac{v}{\|v\|} = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix} \text{ Since } \|v\| = 5$$

Normalize $w \Rightarrow$

$$\frac{w}{\|w\|} = \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix} \text{ Since } \|w\| = 5$$

$$\cos \theta = \frac{v \cdot w}{\|v\| \|w\|} = \frac{3 \cdot 4 + 4 \cdot 3}{5 \cdot 5} = \frac{24}{25}$$

The vector that makes 0° angle with w is $w = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ itself.

A vector that makes 90° angle with w is $\begin{bmatrix} -3 \\ 4 \end{bmatrix}$ or $\begin{bmatrix} 3 \\ -4 \end{bmatrix}$

The vector that make 180° angle with w is $\begin{bmatrix} -4 \\ -3 \end{bmatrix}$.

P5. Solution:

$$(a) V \cdot (-V) = -(V \cdot V) = -\|V\|^2 = -1$$

$$(b) (V+W) \cdot (V-W) = V \cdot V - \cancel{V \cdot W} + \cancel{W \cdot V} - W \cdot W \\ = 1 - 1 = 0$$

$$(c) (V-2W) \cdot (V+2W) = V \cdot V + \cancel{V \cdot (2W)} - \cancel{(2W) \cdot V} - (2W) \cdot (2W) \\ = \|V\|^2 - 4\|W\|^2 \\ = 1 - 4 = -3$$

P7. Solution:

$$(a) W \perp V \Leftrightarrow W \cdot V = 0 \\ \Leftrightarrow 2W_1 - W_2 = 0 \\ W_2 = 2W_1$$

$$\Rightarrow W = \begin{bmatrix} W_1 \\ W_2 \end{bmatrix} = \begin{bmatrix} W_1 \\ 2W_1 \end{bmatrix} = W_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = C \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ \text{for all number } C \text{ (A line)}$$

(b). Plane.

(c). Line.

P12. Solution:

$$(W-CV) \cdot V = 0 \\ W \cdot V - C(V \cdot V) = 0 \\ C = \frac{W \cdot V}{\|V\|^2}$$

For $V = (1, 1)$, $W = (1, 5)$, we have

$$W \cdot V = 6 \\ \|V\|^2 = 2 \\ \Rightarrow C = 3$$

§2.1

P5. Solution: $z = 2$

$$x + y + 6 = 6 \Rightarrow x + y = 0 \\ x - y + 2 = 4 \Rightarrow x - y = 2 \\ \Rightarrow x = 1, y = -1 \\ \text{The point is } (1, -1, 2)$$

$$z = 0$$

$$x + y + 0 = 6 \Rightarrow x + y = 6 \Rightarrow x = 5 \\ x - y + 0 = 4 \Rightarrow x - y = 4 \Rightarrow y = 1$$

The second point is $(5, 1, 0)$

The third point is

$$\frac{(1, -1, 2) + (5, 1, 0)}{2} = (3, 0, 1)$$

P10. Solution:

$$(a) \begin{bmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -2 \\ -4 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} \\ = \begin{bmatrix} 18 \\ 5 \\ 0 \end{bmatrix}$$

$$(b) \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} (2, 1, 0, 0) \cdot (1, 1, 1, 2) \\ (1, 2, 1, 0) \cdot (1, 1, 1, 2) \\ (0, 1, 2, 1) \cdot (1, 1, 1, 2) \\ (0, 0, 1, 2) \cdot (1, 1, 1, 2) \end{bmatrix} \\ = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 6 \end{bmatrix}$$

P17. Solution:

$$(a) I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$(b) P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

P19. Solution: For 2×2 ,

$$E \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y - x \end{bmatrix}$$

let $E = \begin{bmatrix} e_{11} & e_{12} \\ e_{21} & e_{22} \end{bmatrix}$, then.

$$x \begin{bmatrix} e_{11} \\ e_{21} \end{bmatrix} + y \begin{bmatrix} e_{12} \\ e_{22} \end{bmatrix} = \begin{bmatrix} x \\ y - x \end{bmatrix}$$

$$\Rightarrow x e_{11} + y e_{12} = x$$

$$x e_{21} + y e_{22} = y - x$$

$\Rightarrow$ A choice is

$$E = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

For 3×3 , we similarly can get

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$